ALEJANDRO LUIS ARMAS BRAITHWAITE

alarmasb@uchicago.edu | linkedin.com/in/alejandroarmasbr | github.com/Alejandroinfo | alejandroinfo.github.io

EDUCATION

University of Chicago — Chicago, IL
M.S. in Computational Analysis and Public Policy — GPA: 3.59/4.0 — **August 2026**
Relevant Coursework: *Machine Learning, Advanced Statistics, Algorithms, GIS & Spatial Analysis*

Pacífico Business School — Lima, Peru
M.A. in Economics — **December 2019**

Universidad del Pacífico — Lima, Peru
B.A. in Economics — Honors: Top Third of Students in 2015 and 2016 — **December 2017**

PROFESSIONAL EXPERIENCE

The University of Chicago – Policy Lab (with IOM) — Chicago, IL
Policy Lab Consultant | Client: IOM — **August 2025 – December 2025**

- Co-developed (5-person team) a standardized, Sendai-aligned cost model estimating ~US$395M in annual investment needs for gender- and migration-inclusive Disaster Risk Management (DRM) across 5 South American countries.
- Adjusted GDP benchmarks with 4 normalized IDB governance and vulnerability indices, allocated costs across 5 policy dimensions, and validated findings with IOM and UN country offices.

The University of Chicago – Harris School of Public Policy — Chicago, IL
Teaching Assistant – Machine Learning for Public Policy — **August 2025 – December 2025**

- Led TA sessions, held office hours, and graded assignments and quizzes for a section of ~30 graduate students.
- Developed reproducible Python scripts and datasets implementing machine learning methods for public policy applications, supporting model replication and empirical analysis.

Universidad del Pacífico – Faculty of Economics and Finance — Lima, Peru
Lecturer (3 years) | Teaching Assistant (7 years) — **March 2014 – July 2024**

- Taught applied economics and econometrics for 10 years, guiding students through regression analysis with real-world data.

Ministry of Foreign Trade and Tourism (Peru) – COPESCO — Lima, Peru
Public Investment Formulation Assistant — **November 2023 – April 2024**

- Monitored evaluation and compliance for 10+ public tourism infrastructure projects, assessing proposals under Peru's Invierte.pe system.
- Cleaned and structured administrative datasets for feasibility analysis and descriptive statistical reporting.

Ministry of Culture (Peru) – Economics Studies Unit — Lima, Peru
Consultant — **October 2020 – February 2021**

- Consolidated 5 quantitative data sources and qualitative inputs into a centralized Excel dashboard for budget and policy analysis.
- Produced analytical tables and visualizations to support budget allocation decisions and policy briefs.

RESEARCH PROJECTS

Detecting Political Ideology in News Media Articles — Chicago, IL
Research Project, in collaboration with a faculty researcher — **Summer 2025**

- Collected ~60,000 news articles and images (~10K per outlet) from 6 news media outlets using APIs, Selenium, and Requests.
- Cleaned and structured the multimodal (text + image) dataset for downstream NLP and computer-vision classification of political bias.

Chicago Livability Score Project — Chicago, IL
University of Chicago — **Winter 2025**

- Co-designed a weighted livability index for 60 Chicago ZIP codes, building crime, school, and environment sub-indices from 257K records.
- Mapped point-level records to ZIP codes via spatial joins with Census shapefiles (GeoPandas, Shapely) and wrote unit tests (pytest) for a 4-person team's interactive Flask/Folium livability map.

Predicting Dangerous Streets with ML — Chicago, IL
University of Chicago — **Spring 2025**

- Built and compared Logistic Regression, Decision Tree, Random Forest, and Neural Network (MLP) classifiers to predict high-risk NYC street segments, using SMOTE to address class imbalance.
- Found that non-linear models (tuned Random Forest, test F1 0.59 vs. 0.36 random baseline) outperformed logistic regression (0.49), revealing complex interactions among lane count, speed limits, and traffic calming.

TECHNICAL SKILLS

Programming Languages: Python (pandas, scikit-learn, GeoPandas, Shapely, Selenium, Altair), R, Stata, SQL, JavaScript (D3.js)

Software & Tools: Git, Jupyter Notebooks, pytest, Excel (Pivot Tables, VBA, dashboards); **Languages:** English (Fluent), Spanish (Native)

Research Methods: Econometrics & statistical modeling, composite index construction, cost modeling, machine learning, web scraping & API data collection, spatial analysis, reproducible research.